

Universal UAV Propulsion Amperage Estimation Framework

Equation-style mathematical report for small 1404 cinewhoop systems through 9-10 inch multirotor propulsion

$$P_{\text{shaft}} = C_P \rho n^3 D^5$$

Prepared for: **Phone Myat Thu** | GitHub technical documentation and propulsion pre-selection

Important note: This framework estimates electrical current before buying parts or before flight testing. It is not a replacement for safe bench testing, telemetry validation, thermal checks, or local aviation rules.

Source type	Material used in this report	Purpose
Prior technical appendix	High Importance_Amperage Calculation: 2812 1100KV, 24V, 9x4.0 three-blade benchmark	Large-propeller benchmark and previous current-scaling method
Small motor report	SPARK 1404 4000KV test report and datasheet analysis	3-inch high-KV calibration example
Test-data credit	U_FO_UA YouTube channel: https://www.youtube.com/@U_FO_UA	Credit for the uploaded public motor/prop test data source supplied by the user

1. Executive summary

This document replaces plain-text equations with rendered equation blocks matching the mathematical style requested. **All key formulae and numerical results are shown in bold or equation-box form.**

The objective is to produce a repeatable amperage-estimation method that can be used across small high-KV cinewhoop systems, 7-inch FPV systems, and 9-10 inch payload-oriented multirotor systems. The method combines first-principles propeller theory, measured bench-test data, empirical coefficient calibration, and correction factors for real flight.

- **Use measured data first when available.** Datasheet interpolation gives the best estimate for a known motor/prop/voltage combination.
- Use empirical electrical loading coefficient **Psi_e** when RPM is not measured. This lets voltage/current/thrust datasheets become reusable calculation references.
- **Do not transfer constants blindly.** A 3-inch 1404 system and a 9-inch 2812 system obey the same physics, but their coefficient ranges differ due to Reynolds number, blade solidity, pitch, motor loading and RPM regime.
- **Convert bench-test current to flight current using correction factors.** Bench tests are static and usually do not capture frame blockage, propwash recirculation, ducts, wind, battery sag, manoeuvre load, controller corrections and thermal rise.

2. Core notation and unit rules

Symbol	Meaning	Required unit
V	Battery voltage under load	V
I	Motor current or total current	A
P_e	Electrical input power	W
P_shaft	Mechanical shaft power delivered to propeller	W
rho	Air density	kg/m ³ ; use 1.225 at sea-level standard condition
n	Propeller rotational speed	rev/s, not rpm
D	Propeller diameter	m; inch x 0.0254
K_V	Motor velocity constant	rpm/V
lambda	Loaded-speed factor relative to no-load speed	0 to 1
C_P	Propeller power coefficient	dimensionless; true value requires RPM
C_T	Propeller thrust coefficient	dimensionless; true value requires RPM
Psi_e	Empirical electrical loading coefficient	dimensionless-like calibration using no-load speed

Rule: always convert diameter into metres and rpm into revolutions per second before using the aerodynamic formulae.

3. Physics basis: why amperage rises so fast

The foundation is simple: electrical power comes from the battery and becomes motor shaft power after losses. The shaft power then loads the propeller aerodynamically.

$$P_e = V I$$

Electrical input power

$$\text{RPM}_0 = K_V V$$

No-load electrical speed limit

$$n_0 = \frac{\text{RPM}_0}{60}$$

No-load speed in revolutions per second

$$P_{\text{shaft}} = C_P \rho n^3 D^5$$

Propeller shaft-power law

$$T = C_T \rho n^2 D^4$$

Propeller thrust law

$$I = \frac{C_P \rho n^3 D^5}{\eta_e V}$$

Electrical current from propeller loading

Technical meaning: for the same propeller family, current increases approximately with the cube of propeller speed and the fifth power of diameter. This is why a small RPM increase can produce a large current rise.

$$\frac{I_2}{I_1} \approx \left(\frac{n_2}{n_1} \right)^3$$

Cubic current sensitivity

Example: a 10% RPM increase gives $1.10^3 = 1.331$, meaning about **33.1% more current** if the aerodynamic coefficient stays similar.

4. From true C_p to a usable real-world coefficient

True C_P and C_T are excellent academic coefficients, but they require measured RPM. Many hobby test tables provide voltage, current, thrust and power, but not propeller RPM. Therefore, the most practical method is to create an empirical coefficient based on no-load speed.

$$n = \lambda n_0$$

Loaded speed is a fraction of no-load speed

$$I \approx \frac{C_P \rho D^5 \lambda^3 K_V^3 V^2}{\eta_e 60^3}$$

Expanded current model in terms of KV, voltage and diameter

$$\Psi_e = \frac{P_e}{\rho n_0^3 D^5} = \frac{C_P \lambda^3}{\eta_e}$$

Empirical electrical loading coefficient

Psi_e is the practical bridge between datasheets and prediction. It compresses the unknown true propeller coefficient, loaded RPM factor and electrical efficiency into one experimentally calibrated term.

$$P_{e, \text{static}} \approx \Psi_e \rho n_0^3 D^5$$

Static power prediction from empirical loading

$$I_{\text{static}} \approx \frac{\Psi_e \rho n_0^3 D^5}{V}$$

Static current prediction before flight

5. Calibrating Psi_e from known bench data

5.1 Large 9-inch benchmark

Benchmark: 2812 1100KV motor, 6S, 24.0V, 9x4.0 three-blade propeller, measured electrical power 1150-1250W.

$$I_{\min} = \frac{1150}{24.0} = 47.92 \text{ A}$$

$$I_{\max} = \frac{1250}{24.0} = 52.08 \text{ A}$$

$$n_0 = \frac{1100 \times 24.0}{60} = 440 \text{ rev/s}$$

$$D = 9 \times 0.0254 = 0.2286 \text{ m}$$

$$\Psi_{e, 9\text{in}} = \frac{1200}{1.225 \times 440^3 \times 0.2286^5} = 0.01842$$

Result: the 9-inch benchmark gives $\Psi_{e, 9\text{in}} = 0.01765$ to 0.01919 , midpoint 0.01842 .

5.2 Small 1404 4000KV benchmark

Benchmark: SPARK 1404 4000KV at 16V with the GF 3016-3 propeller. The measured full-throttle row gives 11.43A, 368g and 182.88W.

$$n_0 = \frac{4000 \times 16}{60} = 1066.7 \text{ rev/s}$$

$$D = 3.0 \times 0.0254 = 0.0762 \text{ m}$$

$$\Psi_{e, 3016} = \frac{182.88}{1.225 \times 1066.7^3 \times 0.0762^5} = 0.04788$$

Result: GF 3016-3 gives $\Psi_{e, 3016} = 0.04788$. This value is larger than the 9-inch benchmark because the small-prop/high-KV system operates in a very different Reynolds-number and loading regime.

System / propeller	Voltage	KV	D (in)	P _e (W)	I (A)	Thrust	Psi _e
2812 1100KV 9x4x3	24.0 V	1100	9.0	1200.00	50.00	not used here	0.01842
2812 900KV 10x5x3 ref.	22.4 V	900	10.0	969.92	43.30	2700 g user reference	0.01974
1404 4000KV GF 2520-3	16.0 V	4000	2.5	157.76	9.86	316 g	0.10278
1404 4000KV GF D63-3	16.0 V	4000	2.48	198.40	12.40	324 g	0.13447
1404 4000KV GF 3016-3	16.0 V	4000	3.0	182.88	11.43	368 g	0.04788

6. Universal estimation method for pre-purchase calculations

The safest calculation route is to start from measured motor/prop data. When exact data are unavailable, use the closest measured system and scale using the empirical coefficient model.

1. **Estimate all-up mass.** Include frame, battery, flight controller, receiver, camera, VTX, fasteners, prop guards/ducts, payload and wiring.
2. **Choose a design thrust-to-weight ratio.** For calm practice flight, 2:1 may fly; for responsive FPV and payload margin, higher ratios are safer. Do not fly at a design point where hover occurs near maximum throttle.
3. **Find required thrust per motor.** Divide total required thrust by the number of motors.

$$T_{\text{hover, motor}} = \frac{m_{\text{AUW}}}{N_{\text{motors}}}$$

$$T_{\text{design, motor}} = \frac{m_{\text{AUW}} \times \text{TWR}_{\text{design}}}{N_{\text{motors}}}$$

4. **If measured thrust-current points exist, interpolate.** This is more accurate than guessing from KV alone.

$$I(T^*) = I_1 + \frac{(T^* - T_1)(I_2 - I_1)}{T_2 - T_1}$$

5. **If no exact measured row exists, use Psi_e.** Pick the closest data by motor size, KV, voltage, diameter, pitch, blade count and prop family.

$$I_{\text{new}} \approx I_{\text{ref}} \left(\frac{\Psi_{\text{e, new}}}{\Psi_{\text{e, ref}}} \right) \left(\frac{K_{\text{V, new}}}{K_{\text{V, ref}}} \right)^3 \left(\frac{V_{\text{new}}}{V_{\text{ref}}} \right)^2 \left(\frac{D_{\text{new}}}{D_{\text{ref}}} \right)^5$$

7. Pitch, blade count, Reynolds number and frame correction factors

Pitch and blade count are not single fixed multipliers because the response depends on airfoil, chord distribution, blade stiffness, ducting and Reynolds number. For early sizing, use factors conservatively and then replace them with real test data once available.

$$\Psi_{e, new} \approx \Psi_{e, ref} f_{pitch} f_{blade} f_{Re} f_{frame} f_{prop}$$

$$f_{pitch} \approx \left(\frac{(P/D)_{new}}{(P/D)_{ref}} \right)^a \quad a \approx 0.5 \text{ to } 1.2$$

$$f_{blade} \approx \left(\frac{N_{b, new}}{N_{b, ref}} \right)^b \quad b \approx 0.5 \text{ to } 1.0$$

Caution: a two-blade propeller may reduce solidity, but it can also spin faster. Reduced blade count does not always mean lower current at the same throttle command.

$$Re = \frac{\rho v_{tip} c}{\mu}$$

$$V_{tip} = \pi D n$$

$$J = \frac{V_{\infty}}{nD}$$

Small 2.5-3 inch propellers often operate at low Reynolds number and high RPM. Larger 9-10 inch propellers operate with more disk area and usually lower RPM. Therefore, the same equation applies, but the empirical coefficient bands must be separated by propeller class.

8. Static bench-test to real in-flight amperage

Bench tests are static. The propeller works in stationary air on a stand. Real flight changes the inflow, cooling, voltage, frame blockage and control demand. Therefore, convert static current into a flight estimate with correction factors.

$$I_{flight} = I_{static} F_{unload} F_{frame} F_{maneuver} F_{sag} F_{thermal} F_{environment}$$

Factor	Typical planning range	Meaning / when to use
F_unload	0.75-0.95	Forward-flight unloading can reduce current versus static thrust. Hover may be closer to 0.90-1.05.
F_frame	1.00-1.30	Frame blockage, ducts, prop guards, arms and turbulent inflow increase required current. Cinewhoops often need a higher factor.
F_maneuver	1.10-1.60	Control corrections, acceleration, wind recovery, aggressive turns and climb demands.
F_sag	1.00-1.20	Battery voltage sag can push current upward for the same mechanical output.
F_thermal	1.00-1.15	Hot motor windings and ESC losses reduce efficiency as temperature rises.
F_environment	0.95-1.15	Air density changes with altitude, weather and temperature. Lower density requires higher induced velocity and more power.

$$V_{\text{loaded}} = V_{\text{open}} - I R_{\text{total}}$$

$$P_{\text{copper}} = I^2 R_m$$

Thermal consequence: because copper heating is proportional to current squared, reducing current by 15% reduces winding heat to $0.85^2 = 0.7225$, or about 27.8% lower copper heating.

9. Component sizing calculations

$$I_{\text{battery, peak}} = N_{\text{motors}} I_{\text{motor, peak}}$$

$$I_{\text{ESC, continuous}} \geq \frac{I_{\text{motor, peak}}}{U_{\text{ESC}}}$$

$$C_{\text{required}} \geq \frac{I_{\text{battery, peak}}}{\text{Capacity}_{\text{Ah}}}$$

Recommended utilization: for early sizing, use ESC utilization U_{ESC} around 0.70 for continuous margin. This means a motor expected to draw 11.43A peak should be paired with an ESC class of at least 16.33A continuous equivalent, before considering cooling and spikes.

GF 3016-3 example: $I_{\text{motor, peak}} = 11.43\text{A}$, quad peak = 45.72A, ESC continuous screening = $11.43/0.70 = 16.33\text{A}$.

Battery C example: for a 650mAh pack, $C_{\text{required}} = 45.72/0.65 = 70.3\text{C}$ before safety margin. A real battery should be selected with additional margin because advertised C ratings can be optimistic.

10. Worked example A: SPARK 1404 4000KV with GF 3016-3

Use the measured table rather than trying to reuse 9-inch constants. The GF 3016-3 measured data gives 83g at 1.27A and 136g at 2.24A near the low-thrust region.

$$I_{100g} = 1.27 + \frac{(100 - 83)(2.24 - 1.27)}{136 - 83} = 1.58 \text{ A}$$

$$I_{\text{quad, hover}} = 4 \times 1.58 = 6.32 \text{ A}$$

Interpretation: a 400g quad requires about 100g per motor in ideal hover, which interpolates to about 1.58A per motor or 6.32A total static hover current. In real flight, duct/frame losses and manoeuvre demand can push this higher, so plan using a corrected band rather than the raw static value.

AUW	Hover thrust / motor	GF 3016-3 static estimate	Conservative planning band
300 g	75 g	1.12 A/motor; 4.49 A quad	Use roughly 5-8 A depending on frame/duct losses
400 g	100 g	1.58 A/motor; 6.32 A quad	Use roughly 7-11 A depending on frame/duct losses
Full throttle	368 g/motor	11.43 A/motor; 45.72 A quad	Battery/ESC/wiring must tolerate this peak

11. Worked example B: 2812 1100KV, 24V, 9x4.0 three-blade

This example represents a larger propulsion system. The measured static full-throttle current is approximately 47.9-52.1A per motor, with a midpoint of 50.0A.

$$I_{\text{static, mid}} = \frac{1200}{24} = 50.0 \text{ A}$$

$$I_{\text{flight}} = 0.75 \text{ to } 0.85 \times 50.0 = 37.5 \text{ to } 42.5 \text{ A}$$

Important: this 37.5-42.5A band is a parametric in-flight unloading estimate, not a guaranteed measured flight value. Hover, climb, payload, frame blockage and control corrections can move the result up or down.

12. Worked example C: scaling from a 10-inch 2812 reference

Reference: 2812 900KV, 22.4V, 10x5.0 three-blade propeller, 43.3A and about 2700g thrust as a user-provided benchmark.

$$P_e = 22.4 \times 43.3 = 969.9 \text{ W}$$

$$\Psi_{e, 10in} = \frac{969.9}{1.225 \times 336.0^3 \times 0.254^5} = 0.01974$$

Comparison: the 10-inch reference gives $\Psi_{e, 10in} = 0.01974$, close to the 9-inch midpoint 0.01842. That is useful because it suggests the 9-10 inch class has a similar empirical coefficient range when prop style and motor loading are comparable.

13. Adjustment rules: how to reduce current safely

Use the following adjustments when the measured current, motor temperature, ESC temperature, battery sag or flight time is not acceptable.

$$\frac{I_{\text{limited}}}{I_{\text{full}}} \approx \left(\frac{\text{RPM}_{\text{limited}}}{\text{RPM}_{\text{full}}} \right)^3$$

Adjustment	Expected effect	Technical reason
Reduce diameter	Large current reduction	Diameter enters as D^5 , so even small diameter changes matter strongly.
Reduce pitch	Moderate current reduction	Lower pitch reduces blade angle and torque loading.
Reduce blade count	Can reduce current, but not always	Lower solidity reduces drag, but motor may recover RPM. Verify with data.
Lower throttle / motor-output limit	Strong current and heat reduction	Current approximately follows RPM^3 , while heat follows I^2R .
Use lower KV or higher voltage with proper motor selection	Can reduce current for same power	Higher voltage can supply the same power with lower current, but motor/prop matching must be correct.
Reduce AUW	Reduces hover current and improves margin	Required hover thrust is proportional to weight.
Improve cooling and airflow	Reduces thermal risk	Better motor/ESC cooling improves continuous reliability.
Remove frame blockage or duct inefficiency	Reduces flight correction factor	Cleaner inflow reduces wasted power.

Output cap example: if real RPM is reduced to 90% of full RPM, estimated current is about $0.90^3 = 0.729$, or 27.1% lower. If RPM is reduced to 80%, estimated current is $0.80^3 = 0.512$, or 48.8% lower. Throttle percentage is not exactly RPM percentage, so validate with current telemetry.

14. Practical coefficient-selection guide

The table below is not a universal law. It is a practical starting point for early design. Always replace the starting coefficient with real test data when available.

Propulsion class	Suggested starting $\Psi_{e, 10in}$ strategy	Typical caution
2.5-3 inch high-KV cinewhoop / FPV	Use exact datasheet. If unavailable, use the closest 1404/1504/1604 test data and expect $\Psi_{e, 10in}$ to vary widely.	Ducts and guards can increase current significantly.
5-7 inch FPV	Use exact prop/motor test data when possible. Scaling works only within similar KV/diameter/pitch class.	High pitch and aggressive tri-blades can overheat motors quickly.
9-10 inch payload / autonomous quad	Use 9-inch/10-inch measured references around $\Psi_{e, 10in} \approx 0.018\text{-}0.020$ for comparable three-blade loading, then correct for pitch,	Bench current can still be high; ESC and battery margin are critical.

	blade count and voltage.	
Hexa/octo scaling	Calculate per-motor thrust from total AUW and number of motors; do not assume current per motor stays the same if prop/motor changes.	More motors reduce thrust per motor for the same AUW, but add mass and drag.

15. Final before-buying checklist

- **1. Decide AUW and payload margin.** Do not size the propulsion system using empty frame mass only.
- **2. Decide thrust-to-weight target.** Avoid designs that hover near high throttle.
- **3. Find closest measured data.** Match motor size, KV, voltage, prop diameter, pitch and blade count.
- **4. Interpolate current at hover and mission thrust.** Do not use full-throttle current as your hover current.
- **5. Calculate peak current.** Use N_{motors} times per-motor peak current.
- **6. Check ESC margin.** Use continuous rating with cooling and transient headroom.
- **7. Check battery C and voltage sag.** C rating must support peak current without excessive sag.
- **8. Apply flight correction factors.** Ducts, prop guards, payload, wind and controller correction can increase current.
- **9. Bench test safely.** Use guarded tests, short duration, temperature checks and current logging.
- **10. Validate in real flight with telemetry.** Record current, voltage, motor temperature and flight time before assuming the model is correct.

16. Limitations and responsible testing note

The framework is an engineering estimator. It cannot predict exact current without real motor RPM, propeller aerodynamic coefficients, airframe interference, battery internal resistance, ESC timing, motor winding resistance, thermal state and controller behaviour. It should be used to reduce risk and choose reasonable parts before testing, not to skip testing.

Safety note: propulsion testing should be done with proper restraints, guards, eye protection, safe distance, and local regulatory compliance. Do not test high-power propellers in unsafe indoor spaces or with people close to the propeller plane.

17. Source credit and references

Phone Myat Thu. High Importance_Amperage Calculation. University of Hertfordshire technical appendix on 2812 1100KV, 24V, 9x4.0 three-blade current-draw scaling.

Phone Myat Thu. Technical Report: SPARK 1404 4000KV Propeller Current, Thrust and Adjustment Analysis. Datasheet-based analysis for 3-inch FPV / cinewhoop propulsion selection.

U_FO_UA YouTube channel. Public motor/propeller test-data source supplied by the user. URL: https://www.youtube.com/@U_FO_UA

Classical propeller relations used: electrical power $P_e = VI$; propeller shaft power $P_{\text{shaft}} = C_P \rho n^3 D^5$; thrust $T = C_T \rho n^2 D^4$; current estimate $I = P_e/V$.